

Reason About Geometric Relationships and Transformations

Step 9

Upper Elementary

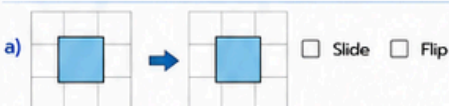


Today I am learning to reason about geometric relationships and transformations.

1. Slide, Flip or Turn?

Look at each pair of shapes.

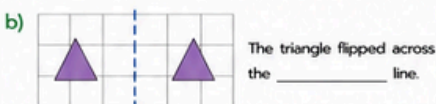
Circle: ☐ Slide ☐ Flip ☐ Turn



2. Describe the Transformation

Look at the picture.

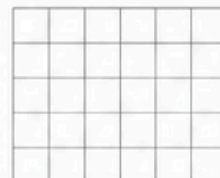
Complete the sentence.



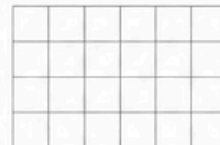
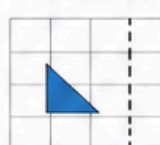
3. Predict the Result

Draw the transformed shape.

- a) Slide the square:
→ 3 spaces right
↑ 1 space up



- b) Flip the triangle across the mirror line.



- c) Turn the arrow:
Quarter turn clockwise.



4. Geometric Relationships

Circle: Yes or No

- a) Do all squares have four right angles? ☐ Yes ☐ No
- b) Are all rectangles quadrilaterals? ☐ Yes ☐ No
- c) Do all triangles have three sides? ☐ Yes ☐ No
- d) Can a square also be a rectangle? ☐ Yes ☐ No

5. Shape Investigation

Complete the table.

Shape	Sides	Corners	Right Angles
Square			
Rectangle			
Triangle			
Pentagon			

6. Transformation Grid

The shape is drawn on the grid.



- a) Slide it 2 squares right.
- b) Flip it across the mirror line.
- c) Rotate it a quarter turn.



7. Real-World Geometry

Look at the objects. Answer the questions.

a) Door



Which shape is it?

b) Window



What geometric properties does it have?

c) Road Sign



What transformations might be used when making identical signs?

8. Think and Explain

- a) What is the difference between a slide and a turn?
- b) How does a flip change a shape?
- c) Why are transformations useful in design and building?



I can...

- ☐ identify slides, flips and turns.
- ☐ describe transformations.
- ☐ reason about geometric relationships.
- ☐ use properties of shapes.
- ☐ apply transformations to solve problems.

Transformation Reminders

Slide



Flip



Turn



Great job!

You are a geometry thinker and a transformation expert!